

Extra Credit Lab - BULK FHIR

- Due Apr 20 by 8am
- Points 0
- Submitting an external tool
- Available Feb 19 at 8am - Apr 20 at 8am

Extra Credit Lab - Bulk FHIR

NOTE: EXTRA CREDIT LABS ARE WORTH UP TO 2 EXTRA POINTS

Overview

This lab will explore FHIR's bulk export functionality. Bulk FHIR helps to retrieve large datasets from a FHIR server which can be used for purposes such as data analysis or export/import at scale. As it does not leverage a FHIR API directly, it instead captures a large volume of resources as an NDJSON file. NDJSON is a format that allows for any number of top level JSON objects rather than requiring a containing JSON structure, with each line being a single JSON object with objects separated by a newline ("\\n" or "\\r\\n").

To complete this lab, you will first explore the Bulk FHIR Interface. Then, you will be provided an example output of FHIR resources in NDJSON and utilize a python notebook to perform simple data analysis and answer the questions given.

Learning Objectives

- Explore BULK FHIR data.
- See how you can perform analysis on bulk FHIR Resources.

Tools Used

- SMARTonFHIR
- [Bulk Export Tool](https://bulk-data.smarthealthit.org/sample-app/index.html?server=https%3A%2F%2Fbulk-data.smarthealthit.org%2FeyJicnliOiIiLCJwYWdlIjoxMDAwMCwiZHVyIjoxMCwidGx0IjoxNSwibSI6MSwic3R1Ijo0LCJkZWwiOjB9%2Ffhir) (<https://bulk-data.smarthealthit.org/sample-app/index.html?server=https%3A%2F%2Fbulk-data.smarthealthit.org%2FeyJicnliOiIiLCJwYWdlIjoxMDAwMCwiZHVyIjoxMCwidGx0IjoxNSwibSI6MSwic3R1Ijo0LCJkZWwiOjB9%2Ffhir>)
- Jupyter Notebook
- [Google Colab](https://colab.research.google.com/) (<https://colab.research.google.com/>)
- [Pretty Print ND JSON](https://pprint-ndjson.neverendingqs.com/) (<https://pprint-ndjson.neverendingqs.com/>)

Resources

- [Bulk Data FHIR IG](https://hl7.org/fhir/uv/bulkdata) (<https://hl7.org/fhir/uv/bulkdata>)

Exercises

Part 1 is purely exploratory and does not need to be submitted anywhere for a grade. For Part 2, you will be submitting to Gradescope.

Part 1 - Exploring the Bulk FHIR Interface

Use the Bulk Export Tool found at <https://bulk-data.smarthealthit.org/sample-app/index.html?server=https%3A%2F%2Fbulk-data.smarthealthit.org%2FeyJicnliOiIiLCJwYWdlIjoxMDAwMCwiZHVyIjoxMCwidGx0IjoxNSwibSI6MSwic3R1Ijo0LCJkZWwiOjB9%2Ffhir> to answer these questions. Open a second tab in your browser and also load <https://pprint-ndjson.neverendingqs.com/> (You may use any method of your choice to view the files, but this is tool is a simple option.)

1. On the Bulk FHIR tool for the given server, there is both a Patient level and System level export option. Switch between the two options and see what changes when you do. How do these differ, and what might be the reasons for the differences?
2. Deselect all resource types but "Device". (There should only be 4 Device resources.) Do not change any other settings so that you do not accidentally filter out needed data. Click the Start Export button. This will bring up the NDJSON file under "Files to Download" at the

bottom. Download your Device export and copy and paste the contents into the Pretty Print ND JSON viewer linked above. Browse through the resources. What are the different types of devices that patients are using, and how many of each device are in use? (**Hint:** The JSON/FHIR path to the field for the name of the device is "Device.type.text".)

- **Note:** As there are only a small number of resources, this should be easily done manually.

3. The Group resource allows a set of members with a common characteristic to be defined, such as a cohort of patients with a particular condition. In the Bulk FHIR tool, select the Group resource (with System-level Export toggled) and download it. Then paste it into your NDJSON viewer. Consider the following about these resources:

- I. What type of resource are the members of all of these Group resources?
- II. While no characteristic is explicitly defined in these particular Group resources, the "name" elements should make it clear what characteristic these groups are intended to capture. What is that characteristic?
- III. What role might a resource such as Group play in bulk export for the sake of analytics? (<https://www.hl7.org/fhir/group.html>) 
(<https://www.hl7.org/fhir/group.html>.)

Part 2 - Data Analysis

Suppose that you are tasked to help out the New England health system to analyze their patient population. They have their data in a FHIR server and you are able to download resources using [FHIR Bulk Export \(https://bulk-data.smarthealthit.org/sample-app/index.html?server=https%3A%2F%2Fbulk-data.smarthealthit.org%2FeyJicnlhOiIiLCJwYWdlIjoxMDAwMCwiZHVyIjoxMCwidGx0IjoxNSwibSI6MSwic3R1Ijo0LCJkZWwiOiJB9%2Ffhir\)](https://bulk-data.smarthealthit.org/sample-app/index.html?server=https%3A%2F%2Fbulk-data.smarthealthit.org%2FeyJicnlhOiIiLCJwYWdlIjoxMDAwMCwiZHVyIjoxMCwidGx0IjoxNSwibSI6MSwic3R1Ijo0LCJkZWwiOiJB9%2Ffhir). To get started, you will need to download the python codebase at <https://github.gatech.edu/cs6440-students/Extra-Credit-Lab-Bulk-FHIR> 
(<https://github.gatech.edu/cs6440-students/Extra-Credit-Lab-Bulk-FHIR>).

The code that you will submit is located in the `bulk\_fhir.eda` module(`src/bulk\_fhir/eda.py`). Follow the instructions on the readme to get your environment setup. You will submit `eda.py` to Gradescope.

To load the dataframe for this assignment, use `get\_bulk\_data` that is imported in `eda.py`. This imports the condition NDJSON file included in `bulk\_fhir.data`.

On the `eda.py` file, there are docstrings for 4 methods that help solve the questions below. Refer to the docstring comments for hints and input, output type information.

1. By using the provided data programmatically determine how many patients have a condition of prediabetes. Use the SNOMED code 15777000. (Note: This is a deprecated SNOMED Code, but it is what appears in the data set. You will not find it in current versions of the SNOMED browser.)
 1. **NOTE:** Follow the return types specified in the function.
2. How many of these patients had prediabetes between the years 1982 and 1990?
3. What cumulative number patients with diabetes we can predict there to be given the past trends by the year 2024? (Create a simple linear regression model that predicts the cumulative number of patients that had a condition of prediabetes over the years.)
 1. **NOTE:** See the eda.py file for more information.
4. Use the SMART Bulk Data exporter to determine the top 3 procedures done for the patient group Health New England. You will have to export the data yourself. (Note: In the case of a tie, you may provide both procedures or select one arbitrarily.)

Submission

Submit just `eda.py` to the assignment in Gradescope.